GRADED QUESTIONS:

1. In this question, you must write a Python program to output the name of the main referee for a given match date (in yyyy-mm-dd format). The input to your program is a file named “date.txt” that has the match date as the first word of the file. Your program must assume that date.txt resides in the same folder as your Python program.

The output name has to be formatted as follows. The last name is displayed followed by the initials of the first name, then a full stop, a space and then the initials of the middle name (if the middle name exists), followed by a full stop.

- For example, if the name of the main referee is “Kennedy Sapam”, the output must be ”Sapam K.”
- If the name of the main referee is “Asit Kumar Sarkar”, the output must be ”Sarkar A. K.”

2. In this question, you must write a Python program to find the cosine of a number obtained from performing a computation on a value retrieved from the database.

1. Find the sum of scores of all host teams satisfying the following conditions.
    a. host_team_score > guest_team_score
    b. name of the host team starts with the character given in the input file ‘parameter.txt’. You have to read the character from the file and use it in your query to retrieve the expected sum. Your program must assume that parameter.txt resides in the same folder as your Python program.
2. Let this sum be denoted by ‘S’. Compute X = S * 10.
3. Assuming that X is a value in radians, convert it into degrees. That is, let X_deg = X * (pi/180).
4. Then, using the *math* library in Python, find *cos*(X_deg) correct up to two decimal places, where *cos* denotes the mathematical trigonometric function *cosine*.
5. For example, if the sum of scores of all host teams satisfying the given conditions is 5, then the output is *round*(*cos*(5*10*(pi/180)),2).

PRACTICE QUESTIONS:

1. In this question, you have to write a Python program to print the names of the players and the team of each player of all those players whose jersey number is a prime number.

1. The list should be ordered in reverse alphabetical order of player names. If two or more players have the same name, then further sorting should be done on the team name, again in reverse alphabetical order.
2. The format of output is as given below:
   Name of the player, followed by a comma (,), then a space and then the team name.

   For example, if Arjun has jersey number 5 and is playing for All Stars and Pranav, with jersey number 7, is playing for team Amigos, then the output will be:
   Pranav, Amigos
   Arjun, All Stars

2. In this problem, you have to write a Python program to print an encoding of the ids of the teams whose jersey colour at home is different from the jersey colour when they play away from home.

- The encoding must be using a shift cipher, which is detailed below.
  - An alphabet is mapped to another alphabet as follows. For a given alphabet α, let *pos* be the position at which α occurs in the alphabet listing (A at 1, B at 2, ... Z at 26). Then the encoding of α is the alphabet at the position (pos + 7) mod 26.
    For example, if M is the alphabet, then the position at which M occurs in the alphabet listing is 13. Then, the encoding of M is the alphabet at the position (13 + 7) mod 26 = 20, which is T.
  - For each digit β, the encoding of β is (β+7) mod 10.
    For example, if 3 is the digit, then the encoding of 3 is the number (3 + 7) mod 10 = 0.
- The ids should be listed in the ascending order before performing the encoding.
- Each line in the output of the program must correspond to one row retrieved from the table.

3. Write a Python program to output the jersey number of the player. Player's name is given in a file named 'player.txt' resides in the same folder as python program file.
The output of the python program is only jersey number.
For example, if the jersey number of the player is 99. Then output must be 99 only.
Note: No spaces.

players		
🔑	player_id	varchar(10)
	name	varchar(80)
	dob	date
	jersey_no	integer
	team_id	varchar(10)

Instructions:

Note: Do not hard code the database name in your program, because your program will be run against a different database instance for evaluation.

For the database connection, use the following connection string variables:

database = sys.argv[1]   //name of the database is obtained from the command line argument

user = os.environ.get('PGUSER')

password = os.environ.get('PGPASSWORD')

host = os.environ.get('PGHOST')

port = os.environ.get('PGPORT')

4. Write a Python program to print the roll number of the student. Student's first name is given in a file named 'name.txt' resides in the same folder as python program file.
The output of the python program is only roll number.
For example, if the first name of the student is 'Vikas'. Then output must be CS01 only.
Note: No spaces.

students	
student_fname	varchar(80)
student_lname	varchar(80)
roll_no	varchar(20)
department_code	varchar(4)
gender	varchar(1)
dob	date
degree	varchar(80)
mobile_no	numeric(10)

Instructions:

Note: Do not hard code the database name in your program, because your program will be run against a different database instance for evaluation.

For the database connection, use the following connection string variables:

database = sys.argv[1] //name of the database is obtained from the command line argument

user = os.environ.get('PGUSER')

password = os.environ.get('PGPASSWORD')

host = os.environ.get('PGHOST')

port = os.environ.get('PGPORT')

5. Write a Python program to print the playground of the given team id. team_id is given in a file named 'team.txt' resides in the same folder as python program file.

• The output of the python program is only playground name.

• For example, if the team_id is 'T0002' . Then output must be Villa Park only

teams		
🔑	team_id	varchar(10)
	name	varchar(80)
	city	varchar(80)
	playground	varchar(80)
	jersey_home_color	varchar(80)
	jersey_away_color	varchar(80)

Instructions:

Note: Do not hard code the database name in your program, because your program will be run against a different database instance for evaluation.

For the database connection, use the following connection string variables:

database = sys.argv[1] //name of the database is obtained from the command line argument

user = os.environ.get('PGUSER')

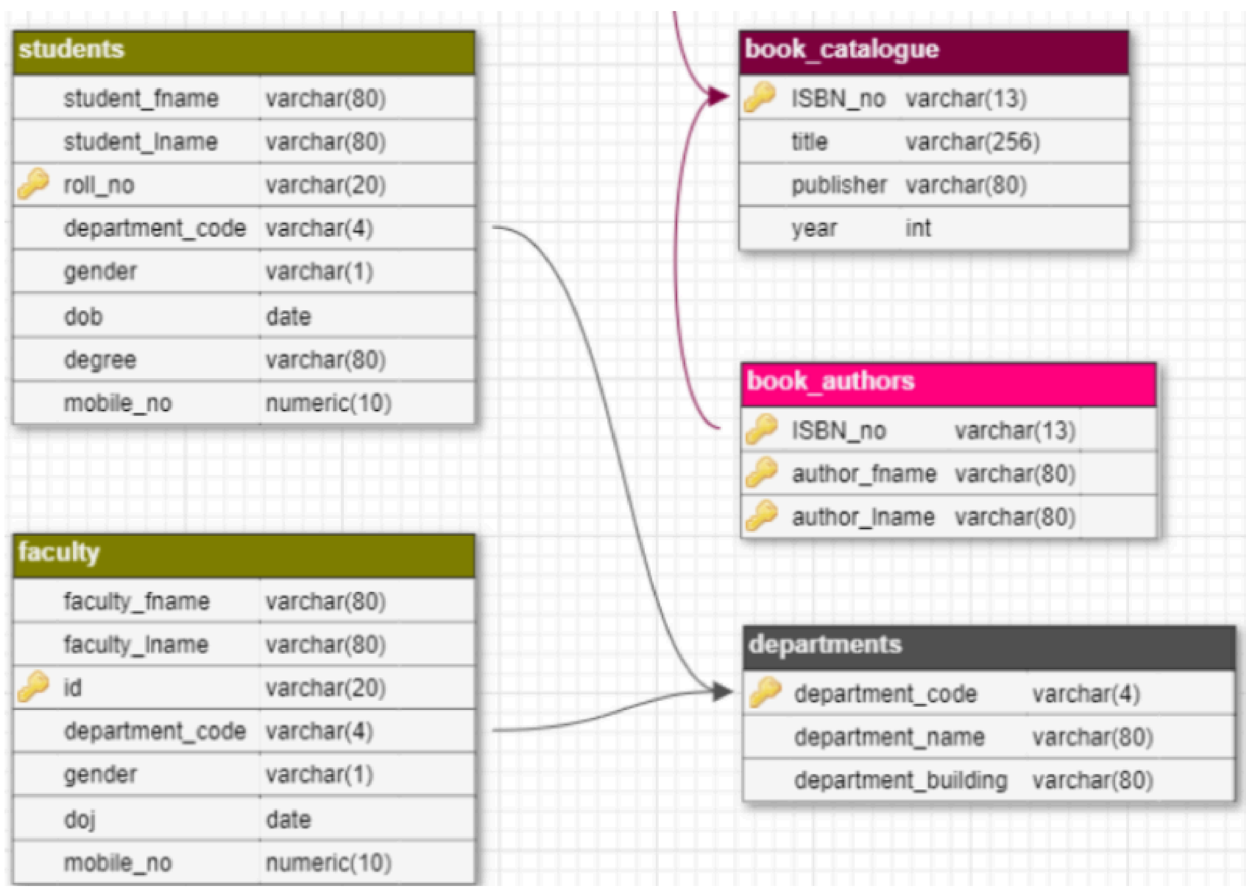
password = os.environ.get('PGPASSWORD')

host = os.environ.get('PGHOST')

port = os.environ.get('PGPORT')

6. Write a Python program to print the student's first name, the corresponding department name and the respective year of date of birth, if the year is even then print "Even" or else "Odd".
Student's first name is given in a file named 'name.txt' resides in the same folder as python program file.

> The output of the python program is only student's first name, the corresponding department name and year of date of birth, if the year is even then "Even" or else "Odd".
> For example, 'Suman' and 'Computer Science' is the name and department name of the student. '2002' is the year he was born in. '2002' is even. Then, the final output will be Suman,Computer Science,Even only. Note: No spaces.
> For example, 'Vinod' and 'Electrical Engineering' is the name and department name of the student. '2003' is the year he was born in. '2003' is not even. Then, the final output will be Vinod,Electrical Engineering,Odd only. Note: No spaces.



Instructions:

Note: Do not hard code the database name in your program, because your program will be run against a different database instance for evaluation.

For the database connection, use the following connection string variables:

database = sys.argv[1]   //name of the database is obtained from the command line argument

user = os.environ.get('PGUSER')

password = os.environ.get('PGPASSWORD')

host = os.environ.get('PGHOST')

port = os.environ.get('PGPORT')